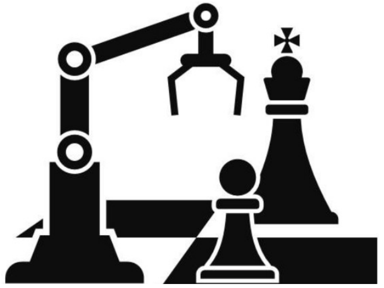
When Intelligence Gets Physical

Mireille El Gheche

June 25, 2026

Writing a novel

Winning Olympic gold medals



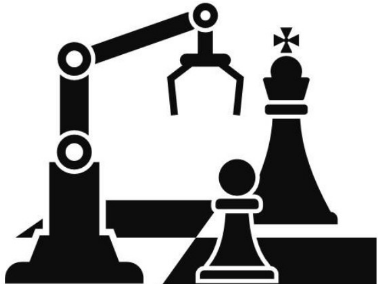
Beating world champions at chess

Operating 24/7 in hazardous environments

What do these achievements have in common?

Writing a novel

Winning Olympic gold medals



Beating world champions at chess

Operating 24/7 in hazardous environments

What do these achievements have in common?

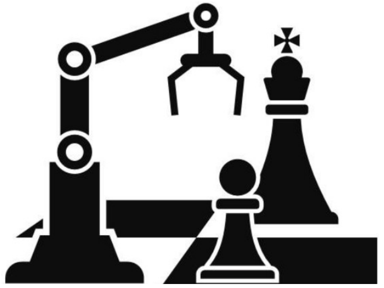
They all require the ability to perceive, learn, reason, adapt, and achieve goals.

Writing a novel

Linguistic, Emotional, Storytelling intelligence

Winning Olympic gold medals

Athletic intelligence



Beating world champions at chess

Computational intelligence

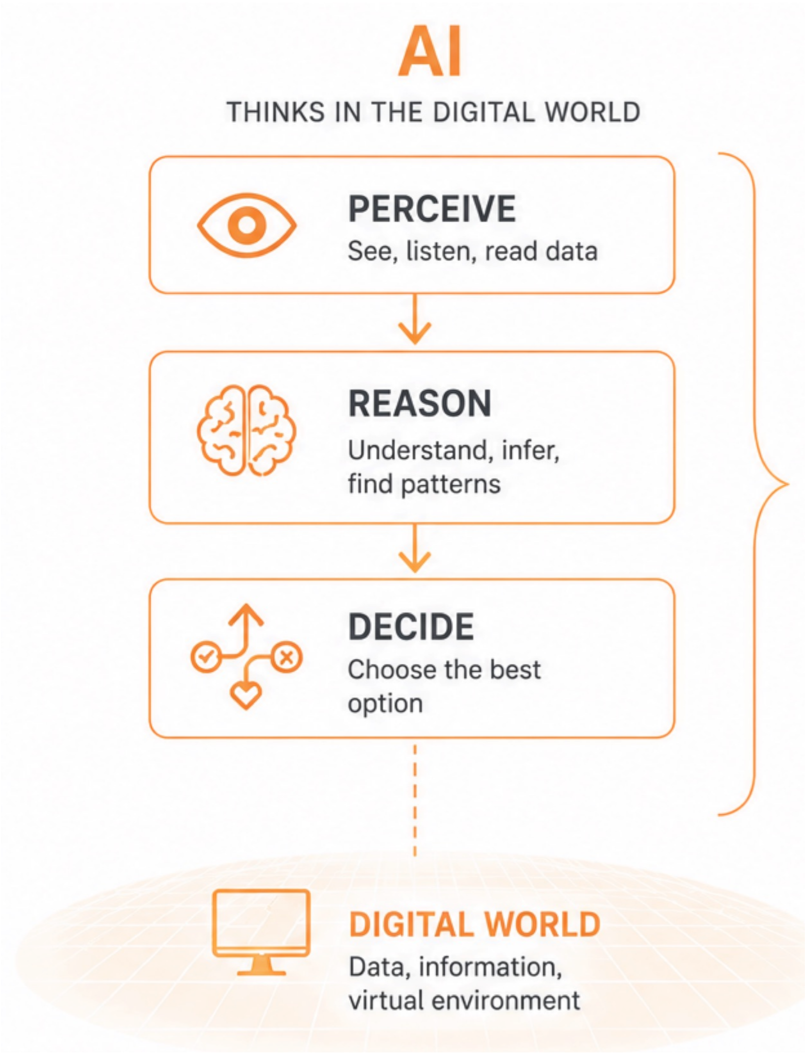
Operating 24/7 in hazardous environments

Physical intelligence

What do these achievements have in common?

Intelligence is not one thing.
Humans, AI systems, and robots are intelligent in very different ways.

When Intelligence Gets Physical


AI
THINKS IN THE DIGITAL WORLD
PERCEIVE
See, listen, read data
REASON
Understand, infer, find patterns
DECIDE
Choose the best option
DIGITAL WORLD
Data, information, virtual environment

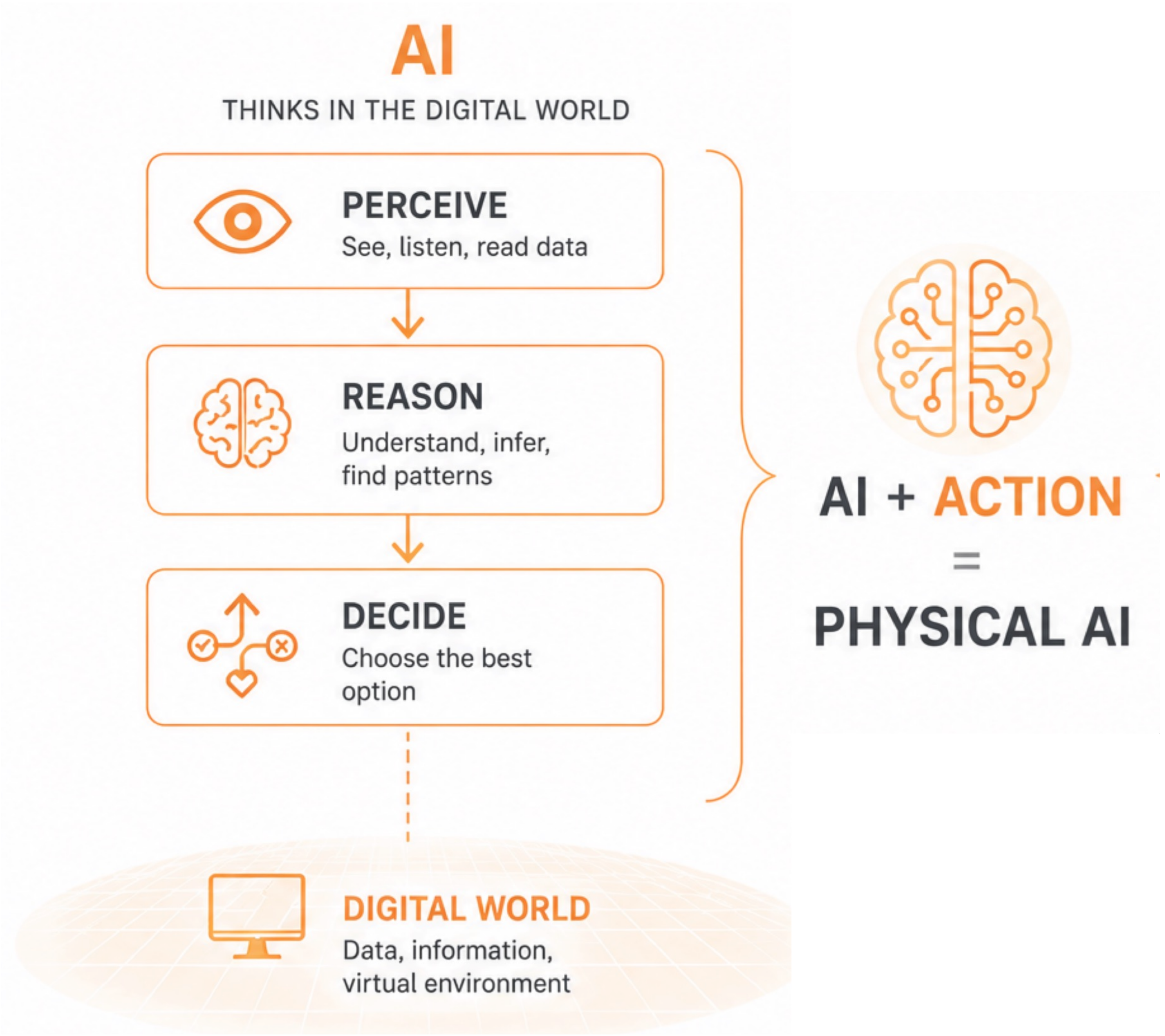
Where AI Already Outperforms Humans?

Deep Blue, 1997

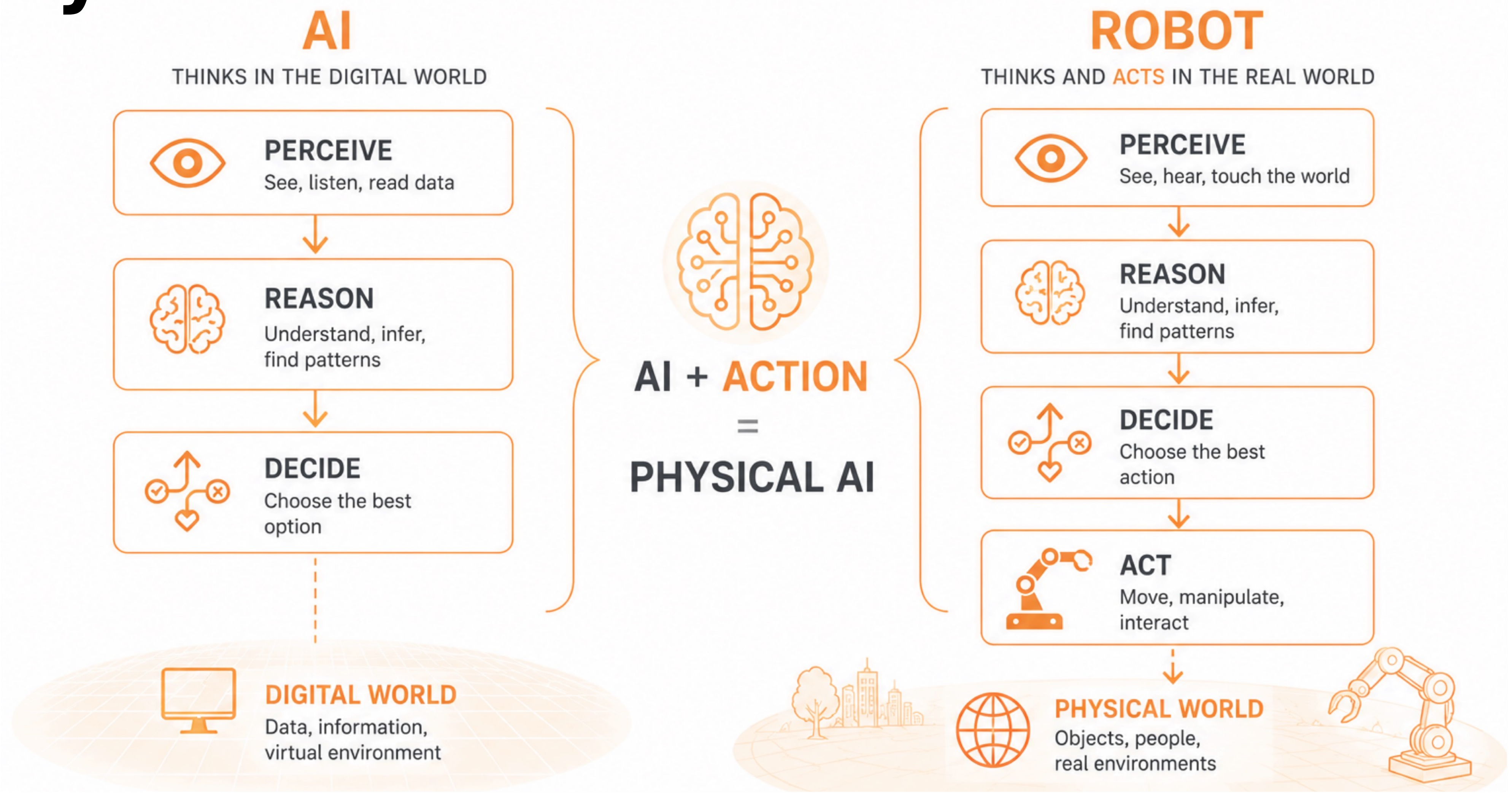
AlphaGo, 2016

GT, 2022

AlphaGenome, 2026


AI
THINKS IN THE DIGITAL WORLD
PERCEIVE
See, listen, read data
REASON
Understand, infer,
find patterns
DECIDE
Choose the best
option
DIGITAL WORLD
Data, information,
virtual environment
AI + ACTION
=
PHYSICAL AI

Why Robotics Is Different?



The physical AI field is where intelligence meets reality.

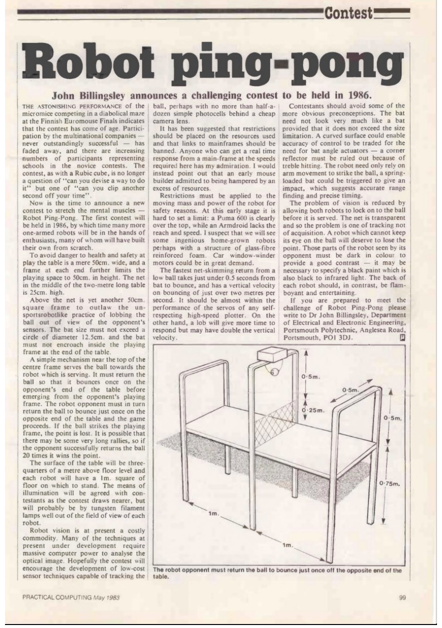
Where Physical AI Outperforms Humans?

ACE, 2026

ACE

What Makes Table Tennis a Grand Challenge for AI?

ACE


Contest

Robot ping-pong

John Billingsley announces a challenging contest to be held in 1986.

THE ASTONISHING PERFORMANCE of the micromice competing in a diabolical maze at the Finnish Euromouse Finals indicates that the contest has come of age. Participation by the multinational companies — never outstandingly successful — has faded away, and there are increasing numbers of participants representing schools in the novice contests. The contest, as with a Rubic cube, is no longer a question of "can you devise a way to do it" but one of "can you clip another second off your time".

Now is the time to announce a new contest to stretch the mental muscles — Robot Ping-Pong. The first contest will be held in 1986, by which time many more one-armed robots will be in the hands of enthusiasts, many of whom will have built their own from scratch.

To avoid danger to health and safety at play the table is a mere 50cm. wide, and a frame at each end further limits the playing space to 50cm. in height. The net in the middle of the two-metre long table is 25cm. high.

Above the net is yet another 50cm. square frame to outlaw the unsportsrobotlike practice of lobbing the ball out of view of the opponent's sensors. The bat size must not exceed a circle of diameter 12.5cm. and the bat must not encroach inside the playing frame at the end of the table.

A simple mechanism near the top of the centre frame serves the ball towards the robot which is serving. It must return the ball so that it bounces once on the opponent's end of the table before emerging from the opponent's playing frame. The robot opponent must in turn return the ball to bounce just once on the opposite end of the table and the game proceeds. If the ball strikes the playing frame, the point is lost. It is possible that there may be some very long rallies, so if the opponent successfully returns the ball 20 times it wins the point.

The surface of the table will be three-quarters of a metre above floor level and each robot will have a 1m. square of floor on which to stand. The means of illumination will be agreed with contestants as the contest draws nearer, but will probably be by tungsten filament lamps well out of the field of view of each robot.

Robot vision is at present a costly commodity. Many of the techniques at present under development require massive computer power to analyse the optical image. Hopefully the contest will encourage the development of low-cost sensor techniques capable of tracking the ball, perhaps with no more than half-a-dozen simple photocells behind a cheap camera lens.

It has been suggested that restrictions should be placed on the resources used and that links to mainframes should be banned. Anyone who can get a real time response from a main-frame at the speeds required here has my admiration. I would instead point out that an early mouse builder admitted to being hampered by an excess of resources.

Restrictions must be applied to the moving mass and power of the robot for safety reasons. At this early stage it is hard to set a limit: a Puma 600 is clearly over the top, while an Armdroid lacks the reach and speed. I suspect that we will see some ingenious home-grown robots perhaps with a structure of glass-fibre reinforced foam. Car window-winder motors could be in great demand.

The fastest net-skimming return from a low ball takes just under 0.5 seconds from bat to bounce, and has a vertical velocity on bouncing of just over two metres per second. It should be almost within the performance of the servos of any self-respecting high-speed plotter. On the other hand, a lob will give more time to respond but may have double the vertical velocity.

Contestants should avoid some of the more obvious preconceptions. The bat need not look very much like a bat provided that it does not exceed the size limitation. A curved surface could enable accuracy of control to be traded for the need for bat angle actuators — a corner reflector must be ruled out because of treble hitting. The robot need only rely on arm movement to strike the ball, a spring-loaded bat could be triggered to give an impact, which suggests accurate range finding and precise timing.

The problem of vision is reduced by allowing both robots to lock on to the ball before it is served. The net is transparent and so the problem is one of tracking not of acquisition. A robot which cannot keep its eye on the ball will deserve to lose the point. Those parts of the robot seen by its opponent must be dark in colour to provide a good contrast — it may be necessary to specify a black paint which is also black to infrared light. The back of each robot should, in contrast, be flamboyant and entertaining.

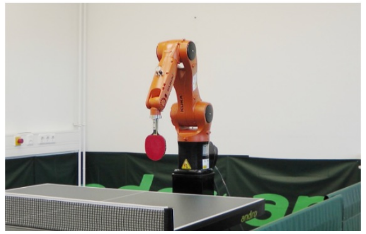
If you are prepared to meet the challenge of Robot Ping-Pong please write to Dr John Billingsley, Department of Electrical and Electronic Engineering, Portsmouth Polytechnic, Anglesea Road, Portsmouth, PO1 3DJ.

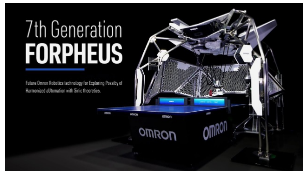
The robot opponent must return the ball to bounce just once off the opposite end of the table.

PRACTICAL COMPUTING *May 1983* 99

[Billingsley et al. 1983]


[Büchler et al. 2020]


[Tebbe et al. 2020]



[Asai et al. 2020]


[Su et al. 2025]

Table Tennis Game:

- Ball speed: up to **32 m/s**
- Extreme spin control: up to **160 rot/s**

Table Tennis Robot:

- Allows for direct comparison of AI with human performance
- Requires very fast perception and decision making in 3D space
- Decades of research. No one had demonstrated head-to-head games with expert human players


[Ambrosio et al. 2024]

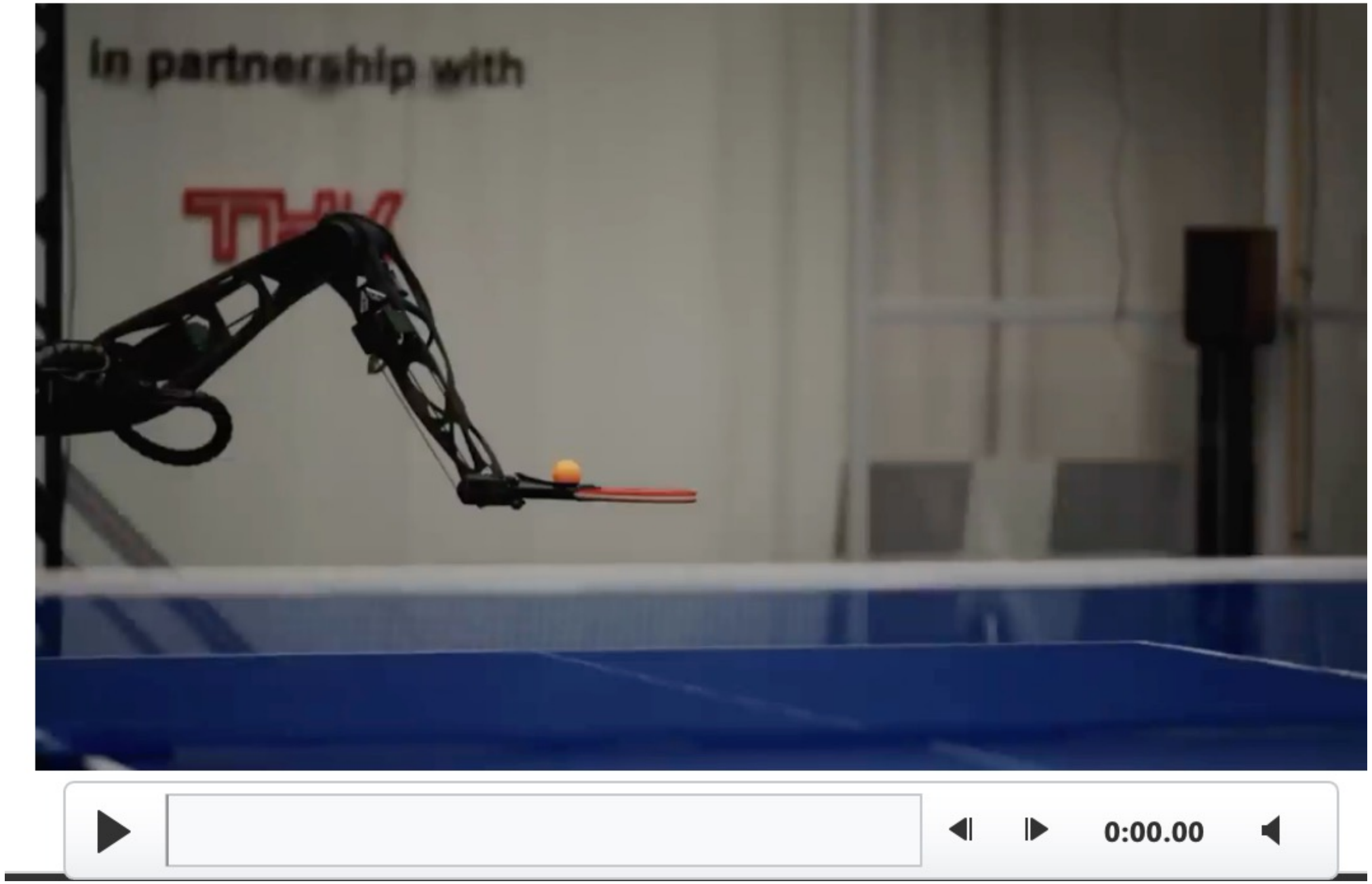
First Competitive Table Tennis Robot

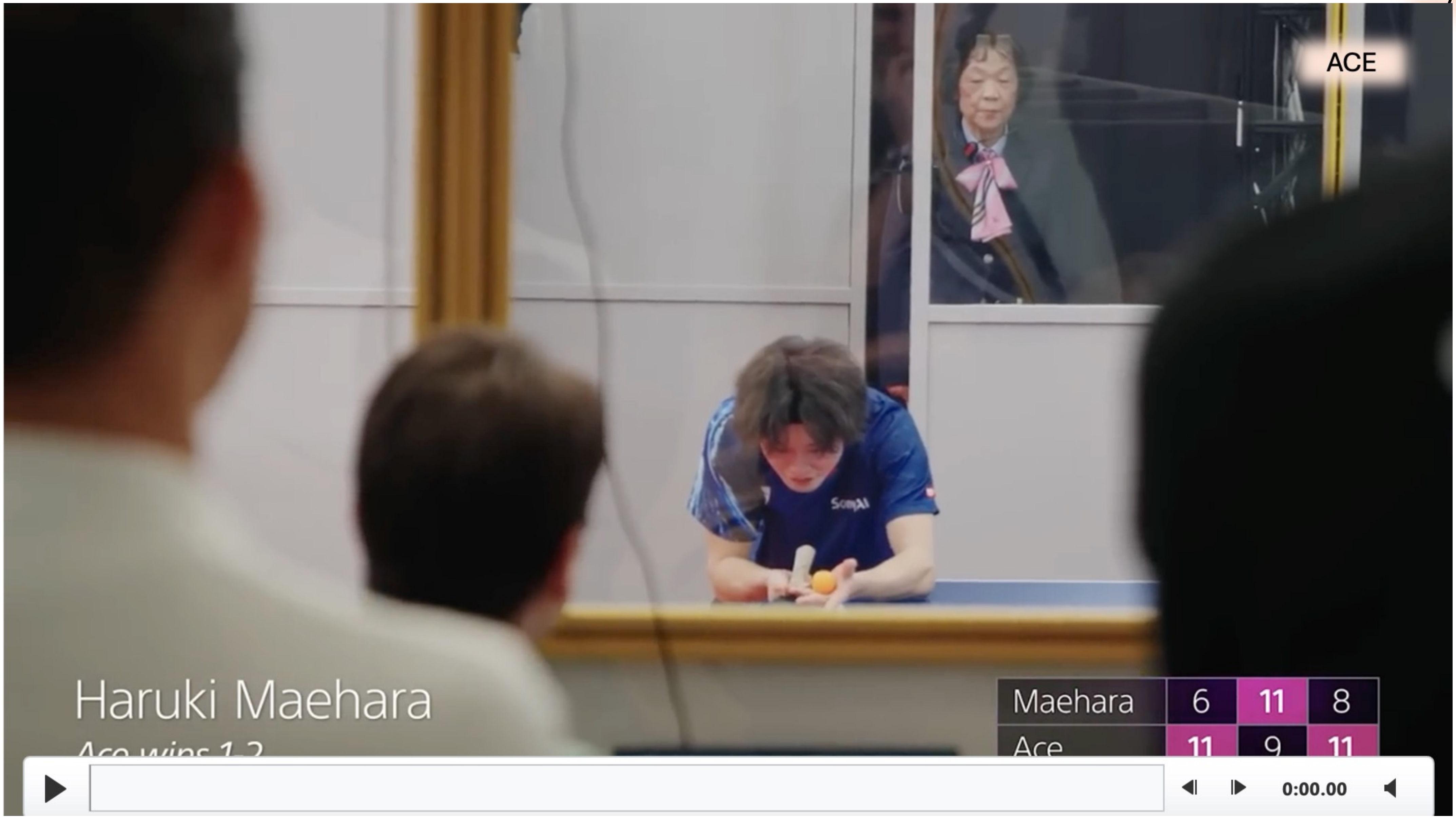
Table Tennis Game:
- Ball speed: up to 32 m/s
- Extreme spin control: up to 160 rot/s

Competitive Games vs. **Elite-level** Opponents including Professional Players

Standard Table Tennis Rules, **Olympic-Size** Court, Licensed Umpires

Standard Equipment (ITTF Certified Balls, Players Use their own racket)




ACE
Haruki Maehara
Maehara 6 11 8
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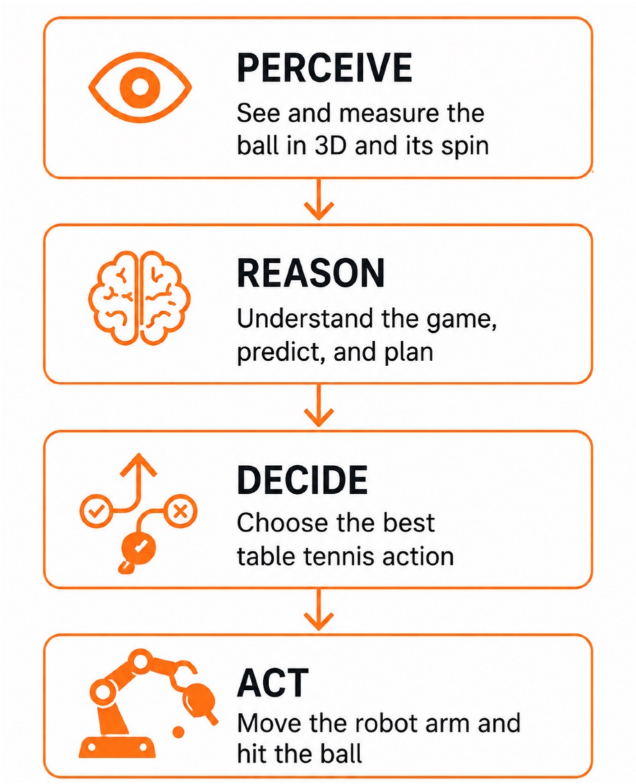
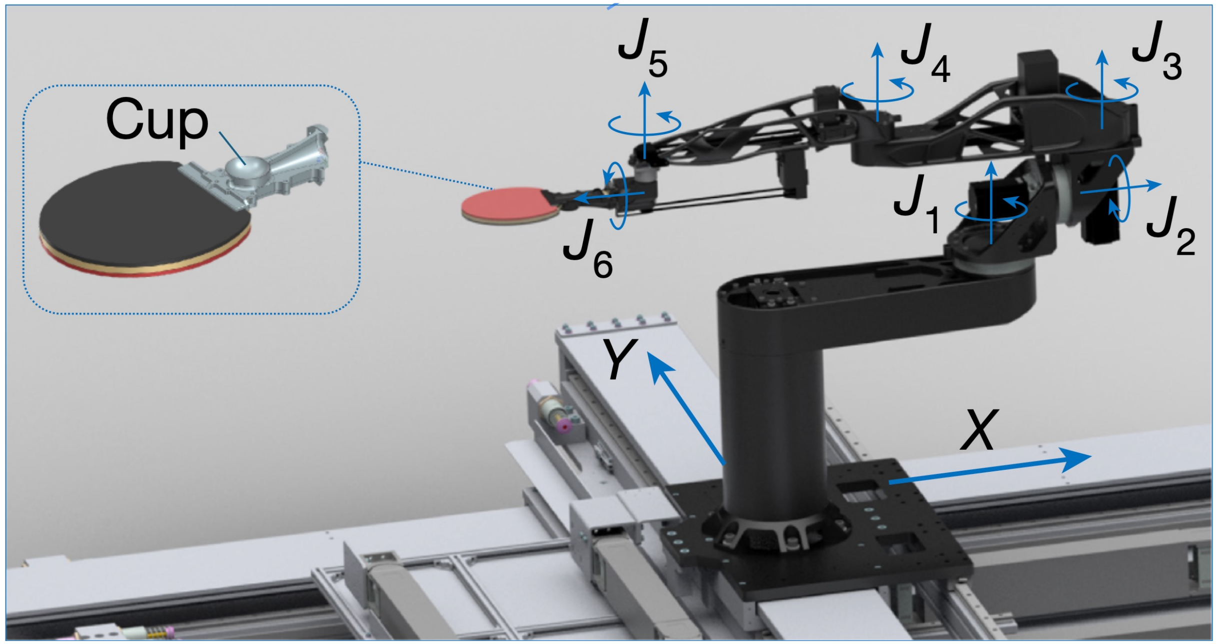
The System



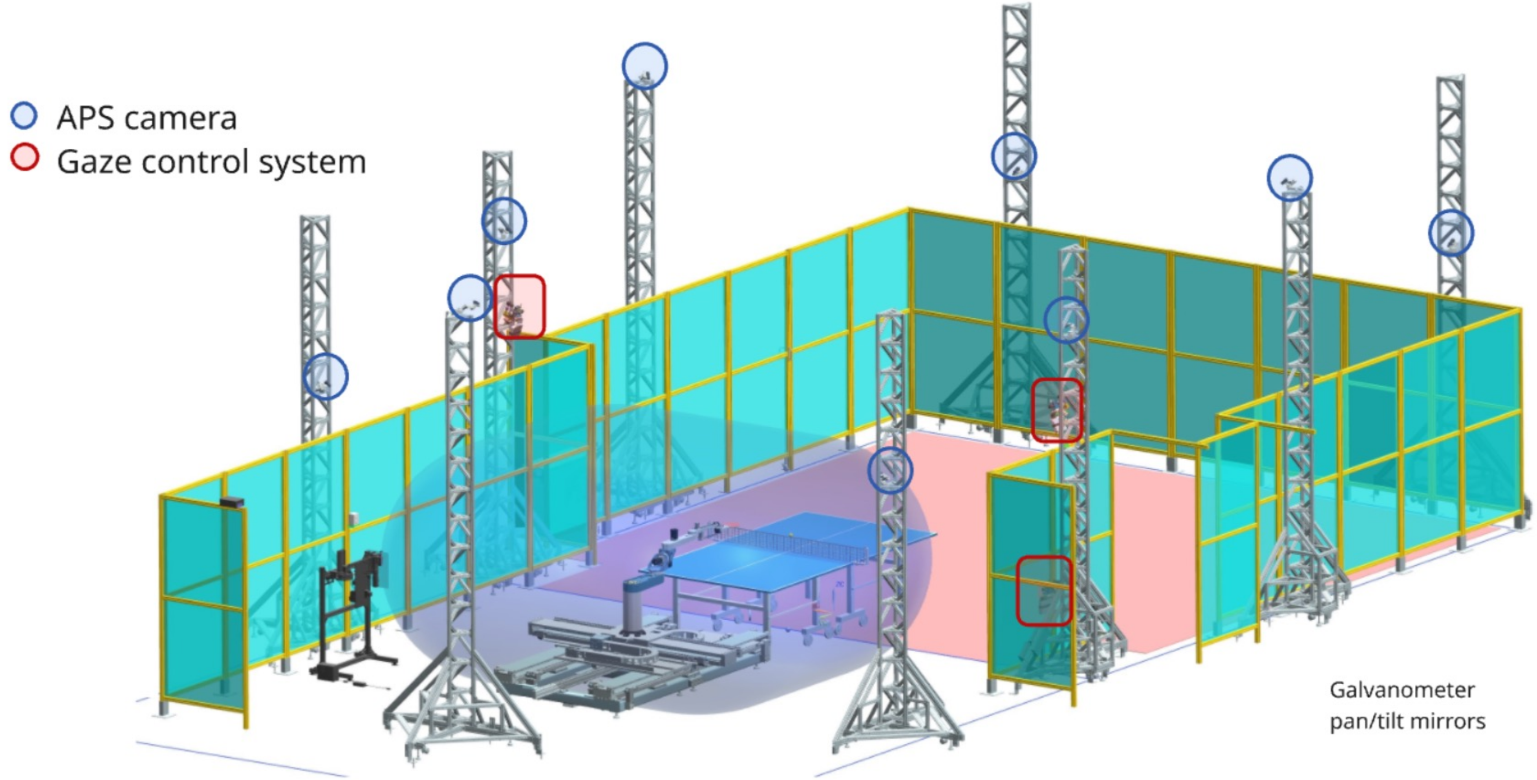


Work area: entire table surface + <2m behind the table

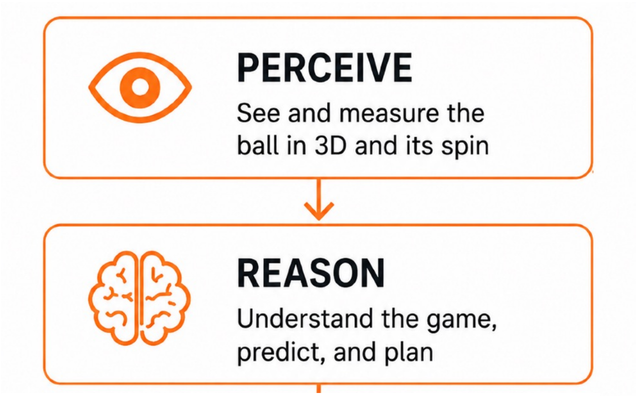
Lightweight & High speed (Over 20m/s full-swing speed)

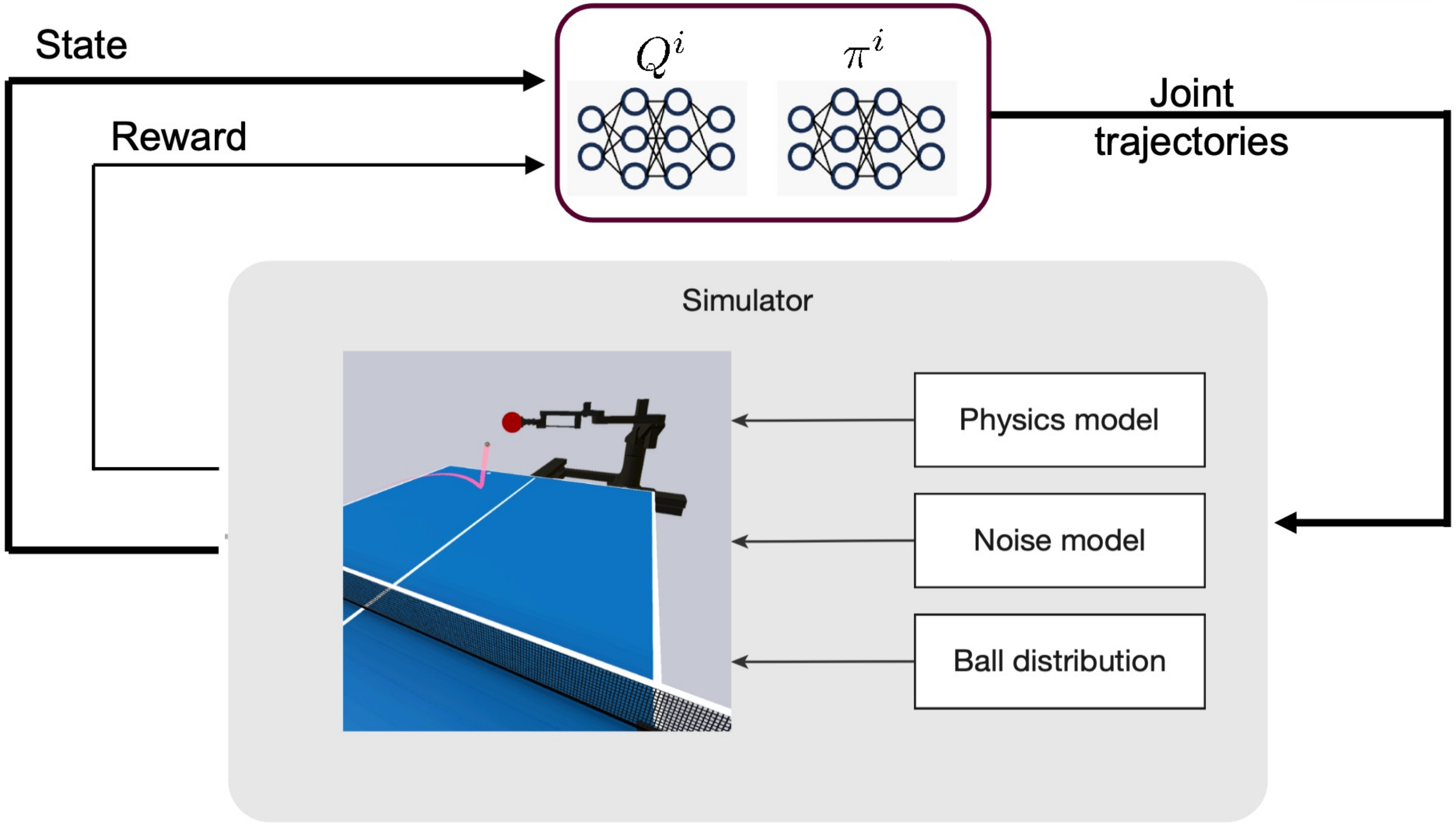
PERCEIVE

See and measure the ball in 3D and its spin



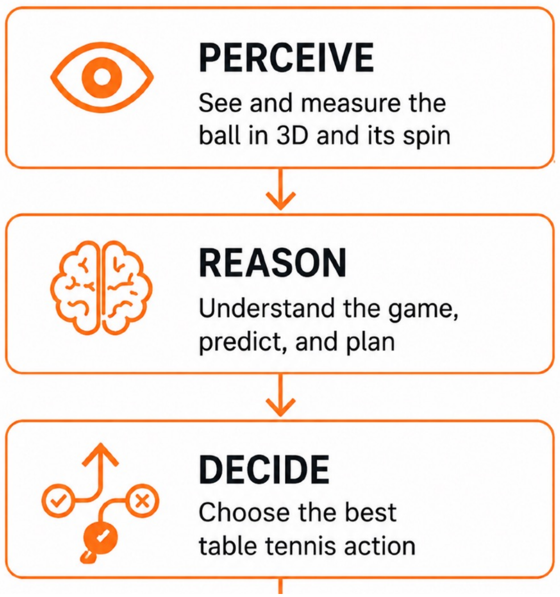
ACE

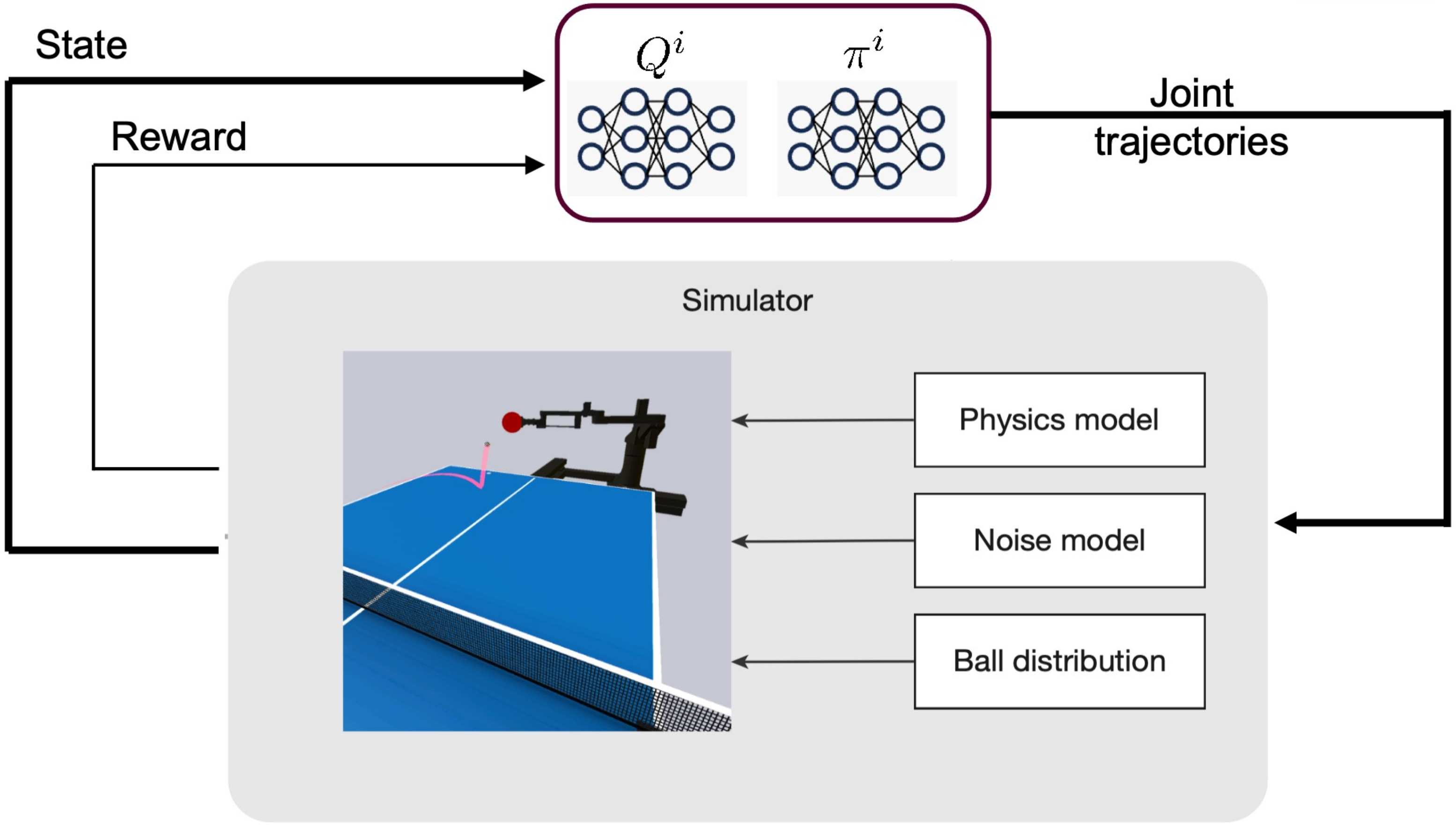




Rally Control Policies Trained via Deep Reinforcement Learning

Robot skills: top spin, back spin, side spin, smash, block…





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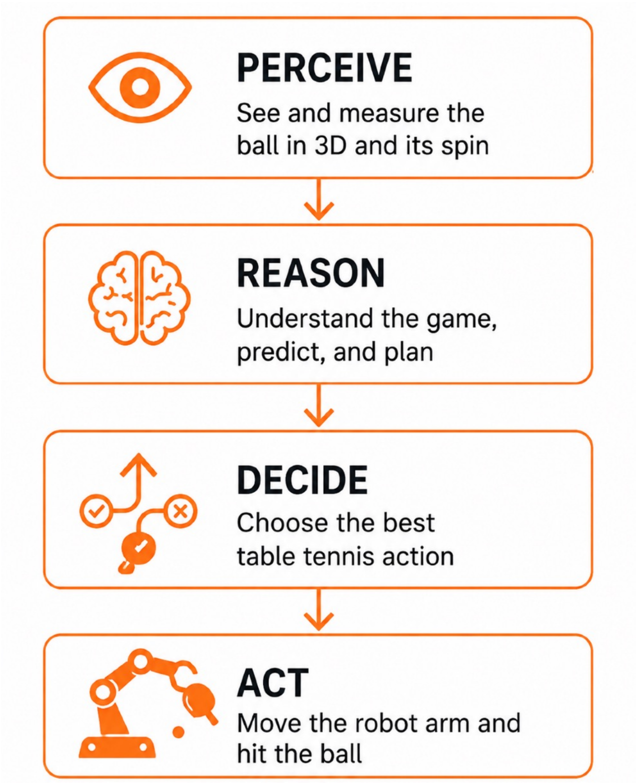

PERCEIVE
See and measure the ball in 3D and its spin
REASON
Understand the game, predict, and plan
DECIDE
Choose the best table tennis action
ACT
Move the robot arm and hit the ball


15:21:53
5
6
Sony AI
ace_r_vs_igarashi_game_1
0:00.00

Robotics is not just a technical field.
It is a human field.

Robots are entering our lives

Moxi

Healthcare
Assist patients & clinicians

Mobility
Move people & goods

Nao

Education
Learning & tutoring

Digit

Industry
Support workers & safety

Care
Support aging populations

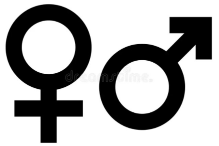
Paro

Daily Life
Everyday services

EmoMime

Why Diversity Matters in Robotics

Who builds the future shapes the future.



Genders

Cultures

Ethnicities

Backgrounds/Experiences/Expertise

Better Perspectives → Better Decisions → Better Robots

Broader Perspectives

Different backgrounds help identify needs, risks, and use cases that may otherwise be overlooked.

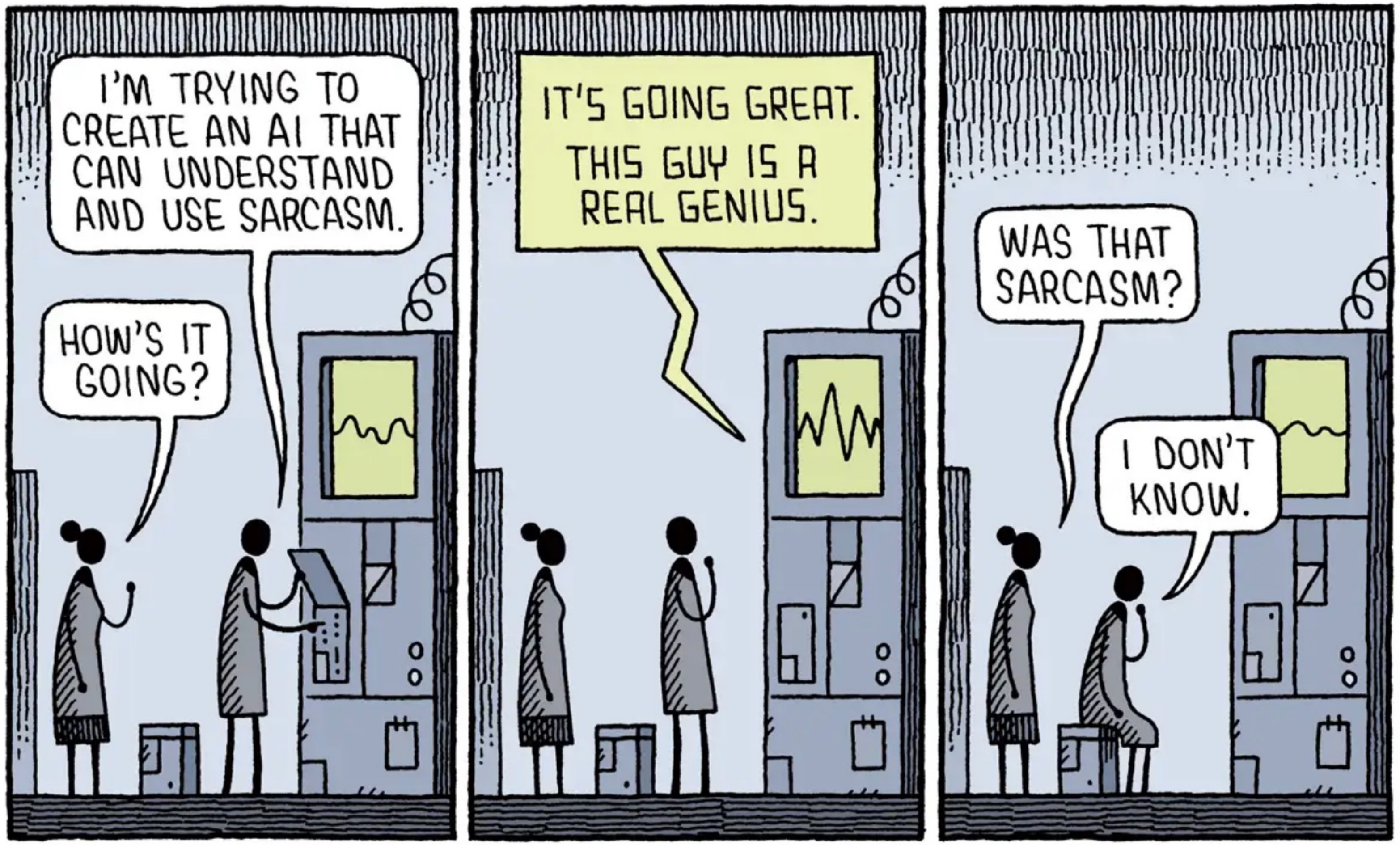
Better Design

Robots are used by diverse people in hospitals, factories, homes, schools, and public spaces.

Better Innovation

Teams with varied expertise and experiences bring different approaches to solving problems.

- AI will become more physical
- Robots will become more collaborative
- Human skills will matter more



The future is not human intelligence versus artificial intelligence.
The future is human intelligence amplified by AI and robotics.

And we all have a role in shaping it.

THANK YOU!